`Extend_2025`: closed-form stoichiometric mixture fraction and closure profile $C(f)$

Mixing-limited reaction closure — `species_limits.py`, `integrate_species_odes`

Setup

Active species $i \in \mathcal{A}$, reactions $j = 1, \ldots, R$. Net stoichiometry matrix

$$N \in \mathbb{R}^{|\mathcal{A}|\times R}, \qquad N_{ij} = \nu_{ij}^{\text{prod}} - \nu_{ij}^{\text{reac}}.$$

Feed vectors $Y^{(1)}, Y^{(2)} \in \mathbb{R}^{|\mathcal{A}|}$ (stream 1 at $f = 1$, stream 2 at $f = 0$). The **no-reaction line** at mixture fraction $f \in [0, 1]$ is

$$M_i(f) = f\, Y_i^{(1)} + (1 - f)\, Y_i^{(2)}.$$

Let y be the current ODE state, y^0 its initial value, and $\langle\cdot\rangle$ the expectation under the step's Beta(α_t, β_t) pdf.

`Extend_2025` shares its selectivity ray with `ray_limit` (Section 1) but replaces `ray_limit`'s general break-point search with a **closed-form** stoichiometric mixture-fraction formula (Section 2) — the classical two-stream f_s formula from reactive-mixing theory, generalized to any number of reactants per stream.

1. Selectivity ray (shared with `ray_limit`)

The accumulated change $\Delta := y - y^0$ lies in range(N). Since $\Delta = N \int_0^t r\, d\tau = N\,\xi(t)$ for the vector of per-reaction extents $\xi_j(t) = \int_0^t r_j\, d\tau' \geq 0$, collapsing the whole network to *one* effective reaction with $d = \Delta$ is exact: Δ *is* $\sum_j \xi_j N_j$, so no extent needs to be tracked or recovered individually — the sum is already the ray. `Extend_2025` reuses `ray_limit`'s own `_selectivity_ray` unchanged:

$$d = \Delta = y - y^0,$$

falling back near y^0 (where $\Delta \approx 0$) to the mass-action projection $d = N\, r(y)$.

The components of d are the coefficients of the **one-line effective reaction**: species with $d_i < 0$ are net reactants with coefficient $\nu_i = -d_i$; species with $d_i > 0$ are net products with coefficient d_i.

2. Closed-form stoichiometric mixture fraction f_s

Rather than the general pairwise-crossing search over all consumed species (`ray_limit`'s `_cmax_along`), `Extend_2025` exploits a structural fact about two-stream feeds: if a net-reactant species is fed **purely** by one stream, its constraint curve $M_i(f)/\nu_i$ is a single line through the origin at that stream's end.

Reduction within a stream. For reactants fed purely by stream 1 ($Y_i^{(2)} = 0$), $M_i(f)/\nu_i = f \cdot (Y_i^{(1)}/\nu_i)$ — a line through $f = 0$. Two such lines with different slopes meet only at $f = 0$, so for $f > 0$ the one with the **smaller** slope is the minimum everywhere on $(0, 1]$: the whole group collapses to a single line with slope

$$\kappa_1 = \min_{i \text{ pure stream 1}} \frac{Y_i^{(1)}}{\nu_i}.$$

Symmetrically, reactants fed purely by stream 2 ($Y_j^{(1)} = 0$) collapse to

$$\kappa_2 = \min_{j \text{ pure stream 2}} \frac{Y_j^{(2)}}{\nu_j}.$$

The crossing. The achievable extent is $c_{\max}(f) = \min\big(f\kappa_1,\ (1-f)\kappa_2\big)$ — one line increasing from 0, one decreasing to 0 — so they cross at exactly one interior point:

$$\boxed{f_s = \frac{\kappa_2}{\kappa_1 + \kappa_2}}, \qquad c^\star = f_s\,\kappa_1 = (1 - f_s)\,\kappa_2.$$

This is the standard two-stream stoichiometric mixture fraction, generalized beyond a single reactant per stream by taking the binding (minimum-ratio) reactant on each side. The complete-reaction limit is the same **two-segment tent** as `ray_limit`'s (only its derivation differs):

$$B_i(0) = Y_i^{(2)}, \qquad B_i(f_s) = M_i(f_s) + c^\star d_i, \qquad B_i(1) = Y_i^{(1)}.$$

Scope. The reduction requires every net-reactant species to be fed by exactly one stream. Species absent from both feeds ($y_i^0 = 0$) can never violate this — since concentrations are non-negative, $d_i = y_i - y_i^0 = y_i \geq 0$, so an unfed intermediate is never a net reactant. If an *actual feed* species is present in both streams and turns out to be a net reactant, the closed form does not apply; `Extend_2025` does not fall back to `ray_limit`'s general search in that case — it prints a diagnostic message and refuses to compute a limit for that step (rates set to 0).

3. Closure profile $C_i(f)$ and scalar λ

Identical machinery to `ray_limit` (same boxed formula, same λ, same segment integrals — only (f_s, B)'s construction differs):

$$\boxed{C_i(f) = B_i(f) + \big[M_i(f) - B_i(f)\big]\,\lambda_i}$$

$$\lambda_i = \mathrm{clip}_{[0,1]}\,\frac{y_i - \langle B_i\rangle}{\langle M_i\rangle - \langle B_i\rangle}.$$

Segment integrals use the regularised incomplete Beta function $I_x(\alpha, \beta)$ (`betainc`):

$$\int_{f_a}^{f_b} (sf + b)\,p_\beta\,df = s\,\frac{\alpha_t}{\alpha_t + \beta_t}\big[I_{f_b}(\alpha_t{+}1, \beta_t) - I_{f_a}(\alpha_t{+}1, \beta_t)\big] + b\,\big[I_{f_b}(\alpha_t, \beta_t) - I_{f_a}(\alpha_t, \beta_t)\big].$$

Because $B_i - M_i = c_{\max}(f)\,d_i$ throughout (same as `ray_limit`), the same **global** λ identity holds: $\lambda_i = 1 - 1/\langle c_{\max}\rangle$ is the same scalar for every species, and $\langle C_i\rangle = y_i$ exactly whenever the clamp is not binding.

4. Worked illustration: `input_fuller_BC_azo_coupling_kinetics`

Scheme: R1: $A{+}B \to R$, R2: $A{+}R \to S$, R3: $A{+}B \to T$, R4: $A{+}T \to S$, R5: $A{+}C \to Q$. **Feeds:** $Y^{(1)} = 15\,A$ (stream 1), $Y^{(2)} = 1.2\,B + 1.2\,C$ (stream 2); $f_{\text{mean}} = 0.0625$.

Same output-step time-midpoint as `ray_limit`'s own worked example ($t = t_{\text{end}}/2 = 0.07629$ s $\approx 3.08\,\tau_s$, $\tau_s = 0.02476$ s, $(\alpha_t, \beta_t) = (0.372,\ 5.575)$ — this instant is $\approx$ 70% A conversion, not 50%), the ray gives the effective reaction

$$0.6556\,A + 0.3314\,B + 0.2868\,C \;\longrightarrow\; 0.2832\,R + 0.0108\,T + 0.0374\,S + 0.2868\,Q.$$

Closed-form f_s. A is pure stream 1 ($Y_A^{(1)} = 15$); B, C are pure stream 2 ($Y_B^{(2)} = Y_C^{(2)} = 1.2$):

$$\kappa_1 = \frac{15}{0.6556} = 22.88, \qquad \kappa_2 = \min\!\left(\frac{1.2}{0.3314}, \frac{1.2}{0.2868}\right) = \min(3.622,\ 4.184) = 3.622 \ (B \text{ binds}).$$

$$f_s = \frac{3.622}{22.88 + 3.622} = 0.1366, \qquad c^\star = f_s\,\kappa_1 = 3.127.$$

Peak values $v_i^{\text{fs}} = M_i(f_s) + c^\star d_i$ (both A and B vanish exactly at f_s, by construction) and the resulting averages and closure scalars:

Species	$\langle M_i \rangle = y_i^0$	v_i^{fs} (mol/L)	$\langle B_i \rangle$ (mol/L)	y_i (mol/L)	λ_i	$\langle C_i \rangle$
A	0.9375	0.000	0.2814	0.2819	0.000839	0.2819
B	1.1250	0.000	0.7934	0.7936	0.000839	0.7936
C	1.1250	0.139	0.8379	0.8382	0.000839	0.8382
R	0.0000	0.886	0.2835	0.2832	0.000839	0.2832
T	0.0000	0.034	0.01076	0.01075	0.000839	0.01075
S	0.0000	0.117	0.03740	0.03737	0.000839	0.03737
Q	0.0000	0.897	0.2871	0.2868	0.000839	0.2868

Every λ_i equals the same global scalar and $\langle C_i \rangle = y_i$ to the quoted digits — identical structure to `ray_limit`. At this same instant `ray_limit`'s own worked example gives $d_A = -0.6556$, $d_B = -0.3314$, $d_C = -0.2868$, $f_s = 0.1366$, $c^\star = 3.127$ — matching to the quoted digits, as expected: A, B, C are each fed by exactly one stream throughout this network (and R, T, S, Q, being unfed, can never violate that condition), so the closed form applies at every step and `Extend_2025` reproduces `ray_limit`'s trajectory numerically, at matching CPU cost, without ever performing the general breakpoint search.

Note: relation to `ray_limit`

`Extend_2025` and `ray_limit` share the selectivity ray d and the entire closure-profile machinery (Section 3); they differ only in how (f_s, B) is constructed from d: `ray_limit`'s search handles *any* feed distribution (including a reactant fed by both streams) via an $O(|C|^2)$ pairwise-crossing search; `Extend_2025`'s closed form handles only the pure single-stream-per-reactant case, in closed form, and refuses outright — rather than silently falling back — when that condition fails. Whenever the condition holds (as it does for every network exercised so far), the two constructions are mathematically identical, not merely similar: both reduce to the unique crossing of one increasing and one decreasing affine function of f.